FORM-2

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COMPLETE

SPECIFICATION

(See section 10 and rule 13)

TITLE OF INVENTION

SYSTEM FOR PREDICTIVE POWER-AWARE ADAPTIVE NEURAL INFERENCE

APPLICANT NAME AND ADDRESS

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THE FOLLOWING SPECIFICATION PARTICULARLY DESCRIBES THE INVENTION AND THE MANNER IN WHICH IT IS TO BE PERFORMED

CROSS-REFERENCE TO RELATED APPLICATIONS AND PRIORITY

[0001] The present application does not claim priority from any patent application.

PREAMBLE

[0002] The following specification particularly describes the invention and the manner in which it is to be performed.

FIELD OF INVENTION

[0003] The present disclosure relates to adaptive neural inference systems and power management for embedded computing devices, and more particularly to a system for predictive power-aware neural inference that dynamically selects and activates pre-optimized neural model configurations based on predicted power states derived from real-time electrical telemetry analysis.

BACKGROUND

[0004] The following text of background art is provided for purposes of understanding and is not an admission that any cited document or technique forms part of the prior art under applicable law.

[0005] Advances in artificial intelligence and machine learning have driven the deployment of intelligent capabilities into diverse applications, including real-time monitoring, autonomous decision-making, industrial automation, wearables, smart cameras, and distributed IoT infrastructures. Traditionally, AI models are trained and executed on cloud servers with abundant compute, memory, and power resources. However, emerging trends in ubiquitous computing and latency-sensitive environments have shifted a significant portion of AI inference workloads toward edge computing devices such as embedded microcontrollers, mobile processors, low-power SoCs, wearable devices, and IoT gateways (IN-202541117248-A).

[0006] These edge devices typically operate under stringent constraints, including limited battery capacity, restricted memory bandwidth, reduced CPU/GPU resources, and variable network connectivity. Deploying conventional AI models on these resource-bound platforms leads to increased energy consumption, degraded responsiveness, or complete infeasibility of operation.

Moreover, many existing model compression techniques—such as pruning, quantization, and knowledge distillation—lack adaptiveness, meaning they are generally applied offline and remain static throughout the device's operational lifetime (IN-202541117248-A).

[0007] Real-world workloads are volatile. Environmental conditions, input data complexity, application priorities, and available battery levels fluctuate continuously. Without dynamic adaptation, an AI model optimized under one operating condition may quickly become inefficient when circumstances change. For example, a vision-based edge sensor may require high-precision inference during events but may shift to lower-energy modes when idle or when battery levels drop (IN-202541117248-A). Existing approaches also fail to incorporate real-time telemetry feedback from the edge device itself. Metrics such as CPU utilization, instantaneous energy draw, temperature, memory availability, and inference latency are rarely used as dynamic optimization triggers for adjusting AI model architecture on the fly (IN-202541117248-A).

[0008] With the growing proliferation of Internet of Things (IoT) applications, edge computing devices have become central to localized, real-time processing. Despite their limitations, there is an increasing demand for deploying complex artificial intelligence (AI) models at the edge to enable smart functionalities such as predictive maintenance, anomaly detection, and real-time personalization. Traditional AI models, designed for powerful cloud infrastructures, are often resource-hungry and not optimized for the edge environment (IN-202511044024-A). Conventional approaches to bring AI to edge devices involve static model compression techniques like pruning, quantization, or knowledge distillation. While these methods achieve notable reductions in model size and inference time, they are often applied offline and lack the capability to adapt dynamically in real-time to fluctuating energy availability or computational resources (IN-202511044024-A).

[0009] Furthermore, given their remote placement, many of the compute devices that perform compute operations may obtain power from solar cells, wind turbines, or other sources that may provide a smaller and less reliable supply of

power than a connection to a power distribution grid. As such, the compute capacity at remote compute locations may fluctuate with the availability of power, leading to an inability to guarantee a fixed level of performance (US-20210004265-A1). The devices located at the edge are often resource constrained and therefore there is pressure on usage of edge resources. Typically, this is addressed through the pooling of memory and storage resources for use by multiple users and devices. The edge may be power and cooling constrained and therefore the power usage may be accounted for by the applications that are consuming the most power (US-20210004265-A1).

[0010] Therefore, there exists a need for a system that not only optimizes AI models for edge devices but does so adaptively, based on real-time operational conditions, device constraints, and application-level requirements.

[0011] The present invention addresses the above shortcomings of the prior art. However, the invention is entirely different from the prior art in terms of novelty and technological advancements.

OBJECT

[0012] In view of the foregoing, and without prejudice to the generality of the disclosure, it is an object of the present invention to provide. It is to be understood that the foregoing objects are illustrative and non-limiting; additional objects will be apparent from the description, examples, and claims, and need not be achieved in every embodiment.

[0013] The object of the present disclosure is to provide a neural inference system with predictive power awareness that can maintain steady operation in the face of erratic or fluctuating electrical conditions. The system aims to continuously monitor electrical input characteristics to extract forecast signals for approaching voltage instability, thereby enabling proactive adaptation rather than reactive responses to power degradation.

[0014] Another object of the present disclosure is to establish a predictable link between neural network execution configuration and power availability trends. By developing a deterministic power-to-model mapping mechanism, the system seeks to enable precise scaling and dynamic neural model selection in response to

anticipated power constraints, ensuring that reconfiguration decisions are repeatable and methodical rather than arbitrary.

[0015] A further object of the present disclosure is to enable continuous inference operation in the case of brownouts or brief voltage drops. The system aims to adaptively lower energy and computing demands in response to fluctuating power supply by dynamically switching between pre-optimized neural model configurations of varying computational complexity, including full-precision, quantized, and lightweight configurations.

[0016] Yet another object of the present disclosure is to use a closed-loop feedback mechanism to bring together neural network architecture control with electrical system telemetry. This integration aims to transform edge intelligence from static deployment to predictive and energy-adaptive execution, thereby improving operational resilience and durability in electrically unstable edge computing environments.

[0017] For the avoidance of doubt, the foregoing objects encompass compositions and manufacturing processes suitable for adoption in jurisdictions where claims to methods of treatment of animals may be restricted; corresponding "use" or "product" claim formats are contemplated without limitation.

SUMMARY

[0018] The following summary is intended to introduce aspects of the present invention in a simplified form and does not identify all the features or the scope of the claimed invention. The summary is provided to assist in understanding the invention and is to be read in conjunction with the detailed description and accompanying drawings, claims, and abstract. One or more embodiments of the present disclosure are set forth below solely for illustrative purposes and should not be construed as limiting.

[0019] According to an aspect of the present disclosure, a system (100) for predictive power-aware neural inference is provided. The system (100) comprises an electrical telemetry module (110) configured to continuously sample an input voltage signal from a voltage input source (102) and extract derived electrical indicators from the sampled voltage signal. The derived electrical indicators include

at least a rate of change of voltage and a short-term variance. The system (100) further comprises a predictive control engine (120) communicatively coupled to the electrical telemetry module (110). The predictive control engine (120) is configured to receive the derived electrical indicators from the electrical telemetry module (110), compute a predictive power stability index based on the derived electrical indicators, and determine a predicted power state based on the predictive power stability index. The system (100) additionally comprises an adaptive neural execution layer (130) communicatively coupled to the predictive control engine (120). The adaptive neural execution layer (130) is configured to store a plurality of pre-optimized neural model configurations having different computational complexities and dynamically select and activate one of the plurality of pre-optimized neural model configurations based on the predicted power state to maintain continuous neural inference.

[0020] The technical advantages associated with the foregoing features include enabling proactive adaptation of neural inference operations in response to anticipated power conditions rather than reactive responses to power failures. By extracting derived electrical indicators including rate of change and short-term variance, the system can detect early warning signs of voltage instability before degradation occurs. The predictive power stability index provides a quantitative basis for model selection decisions, enabling systematic and repeatable reconfiguration. The dynamic selection and activation of pre-optimized neural model configurations allows the system to maintain continuous inference operation even during brownouts or voltage fluctuations, thereby improving operational resilience in electrically unstable edge computing environments.

[0021] According to other aspects of the present disclosure, the derived electrical indicators may further include an instantaneous voltage magnitude and a moving average voltage computed over a sliding time window.

[0022] The technical advantages associated with the foregoing features include providing a more comprehensive characterization of the power supply state. The instantaneous voltage magnitude captures immediate voltage levels, while the moving average voltage computed over a sliding time window filters out transient

noise and provides a smoothed representation of voltage trends. Together, these additional indicators enhance the accuracy of power state prediction and enable more informed model selection decisions.

[0023] According to other aspects of the present disclosure, the plurality of pre-optimized neural model configurations may comprise a full-precision neural model configuration, a reduced-precision quantized neural model configuration, and a computationally lightweight neural model configuration.

[0024] The technical advantages associated with the foregoing features include providing a graduated range of model options that can be matched to varying power availability conditions. The full-precision neural model configuration delivers maximum inference accuracy when power is stable and abundant. The reduced-precision quantized neural model configuration offers a balance between accuracy and energy consumption during moderate power instability. The computationally lightweight neural model configuration enables continued inference operation with minimal energy demands during severe power constraints, thereby ensuring uninterrupted service across a spectrum of power conditions.

[0025] According to other aspects of the present disclosure, the predictive control engine (120) may be further configured to implement a deterministic power-to-model mapping function that maps the predicted power state to a corresponding neural model configuration from the plurality of pre-optimized neural model configurations.

[0026] The technical advantages associated with the foregoing features include establishing a predictable and repeatable relationship between power conditions and model selection. The deterministic power-to-model mapping function ensures that identical power states result in identical model selections, enabling consistent system behavior and facilitating debugging, testing, and validation of the adaptive inference system.

[0027] According to other aspects of the present disclosure, the predictive control engine (120) may be further configured to apply hysteresis and state persistence constraints to prevent rapid oscillations between neural model configurations during fluctuating power conditions.

[0028] The technical advantages associated with the foregoing features include improving system stability and reducing computational overhead associated with frequent model switching. The hysteresis and state persistence constraints prevent the system from oscillating rapidly between different model configurations when power conditions hover near transition thresholds, thereby reducing switching-related latency and ensuring smoother inference operation.

[0029] According to other aspects of the present disclosure, the predictive control engine (120) may comprise a long short-term memory (LSTM) model configured to analyze a time-series sequence of voltage readings and classify a current power condition as one of a plurality of power states.

[0030] The technical advantages associated with the foregoing features include leveraging the temporal pattern recognition capabilities of LSTM networks to identify power condition trends from historical voltage data. The LSTM model can learn complex temporal dependencies in voltage sequences, enabling more accurate prediction of impending power state changes compared to instantaneous threshold-based approaches.

[0031] According to other aspects of the present disclosure, the adaptive neural execution layer (130) may be further configured to perform low-latency switching between the plurality of pre-optimized neural model configurations by preloading the neural model configurations in memory.

[0032] The technical advantages associated with the foregoing features include minimizing the time delay associated with transitioning between different neural model configurations. By preloading the neural model configurations in memory, the system avoids the latency that would otherwise be incurred by loading models from storage during power state transitions, thereby enabling rapid adaptation to changing power conditions and maintaining inference continuity.

[0033] According to other aspects of the present disclosure, the predictive power stability index may comprise a weighted combination of a moving average voltage, the rate of change of voltage, and the short-term variance.

[0034] The technical advantages associated with the foregoing features include providing a composite metric that captures multiple dimensions of power supply

behavior. The weighted combination allows the system to balance the contributions of different electrical indicators according to their relevance for predicting power stability, enabling nuanced assessment of power conditions that accounts for both current voltage levels and voltage dynamics.

[0035] According to other aspects of the present disclosure, the adaptive neural execution layer (130) may be further configured to implement a state retention mechanism to maintain continuity of an intermediate inference context during switching between neural model configurations.

[0036] The technical advantages associated with the foregoing features include preserving computational progress and contextual information when transitioning between neural model configurations. The state retention mechanism enables the system to resume inference from an intermediate state rather than restarting from the beginning, thereby reducing redundant computation and maintaining inference coherence across model transitions.

[0037] According to other aspects of the present disclosure, the predictive control engine (120) may be further configured to generate a normalized predictive power stability score for uniform interpretation across different hardware platforms and voltage ranges.

[0038] The technical advantages associated with the foregoing features include enabling portability and consistent behavior of the adaptive inference system across diverse deployment environments. The normalized predictive power stability score abstracts away hardware-specific voltage characteristics, allowing the same power-to-model mapping logic to be applied uniformly regardless of the underlying hardware platform or nominal voltage range.

[0039] The foregoing general description of the illustrative embodiments and the following detailed description thereof are merely exemplary aspects of the teachings of this disclosure and are not restrictive.

BRIEF DESCRIPTION OF FIGURES

[0040] The drawings appended herein illustrate exemplary embodiments of the present invention and are provided to enhance understanding of the inventive features disclosed. These figures, when read in conjunction with the detailed

description, depict the operational flow, component architecture, and other mechanisms.

[0041] The illustrations are schematic in nature and serve solely to clarify the principles and functionality of the present disclosure. While particular constructions and configurations are depicted for explanatory purposes, it will be understood that the invention is not limited to the precise implementations shown in the figures and may encompass variations without departing from the scope of protection.

[0042] FIG. 1 illustrates a block diagram of a system for predictive power-aware neural inference, according to aspects of the present disclosure.

[0043] FIG. 2 illustrates a state transition diagram of adaptive neural model switching governed by predicted power conditions, according to an embodiment.

[0044] FIG. 3 illustrates a schematic of an electrical telemetry module and a predictive control engine, according to aspects of the present disclosure.

[0045] FIG. 4 illustrates a schematic of an adaptive neural execution layer with a pre-optimized model ensemble, according to an embodiment.

[0046] FIG. 5 illustrates a flowchart of a method for predictive power-aware neural inference, according to aspects of the present disclosure.

[0047] Furthermore, in terms of the construction of the system, one or more components of the device may have been represented in the figures by conventional symbols, and the figures may show only those specific details that are pertinent to understanding the embodiments of the present invention so as not to obscure the figures with details that will be readily apparent to those of ordinary skill in the art having benefit of the description herein.

DETAILED DESCRIPTION

[0048] The following detailed description is intended to provide an in-depth understanding of the invention and its various components, configurations, and operational aspects. While specific embodiments are described with reference to the accompanying drawings, the invention is not limited to the particular forms disclosed herein. The scope of the invention shall be interpreted broadly and in accordance with the claims appended hereto.

[0049] The terms “comprises,” “comprising,” “includes,” “including,” and similar expressions used in this description are intended to be open-ended and should be interpreted to include, but not be limited to, the mentioned elements. Unless explicitly stated otherwise, singular terms may include their plural equivalents and vice versa, depending on the context in which they appear.

[0050] Reference is now made to the accompanying figures, which illustrate example embodiments of the present invention. It should be appreciated that the figures are intended for illustration and explanatory purposes only and do not limit the scope of the invention. Similar reference numerals have been used to denote functionally similar components throughout the figures.

[0051] The embodiments described herein are presented for illustrative purposes and are subject to variations, modifications, and adaptations by those skilled in the art. Any equivalent implementations that perform substantially the same function in substantially the same manner are intended to fall within the scope of this disclosure and the claims appended hereto.

[0052] The use of the term “exemplary” in the description shall be understood to mean “serving as an example or illustration” and shall not be construed as limiting the invention to the preferred embodiments disclosed. Various other embodiments may be developed without departing from the spirit or essential characteristics of the invention. The technical principles and advantages outlined in the foregoing description should be read in conjunction with the figures and claims. It is expressly understood that while the detailed description sets out specific structural and operational aspects, the scope of protection is governed by the claims and includes all legal equivalents of the claimed subject matter. Embodiments of the disclosure are described in the following paragraphs with reference to Figures. In Figures, the same elements or elements that have the same functions are indicated by the same reference signs.

[0053] The present invention will now be described in detail with reference to the accompanying drawings illustrating exemplary embodiments.

[0054] FIG. 1 illustrates a block diagram of a system 100 for predictive power-aware neural inference according to various embodiments. The system 100 may be

deployed in an edge device placed in an environment with an electricity supply that fluctuates. The system 100 comprises an electrical telemetry module 110, a predictive control engine 120, and an adaptive neural execution layer 130 that are interconnected to form a closed-loop framework for predictive and energy-adaptive neural execution.

[0055] The system 100 receives electrical power from a voltage input source 102. The voltage input source 102 may comprise various power supply types including, but not limited to, batteries, solar panels, wind turbines, grid connections, fuel cells, or other power supplies. In some cases, the voltage input source 102 may provide a smaller and less reliable supply of power than a connection to a power distribution grid, such as when the system 100 is deployed at remote compute locations.

[0056] The electrical telemetry module 110 is configured to continuously sample an input voltage signal from the voltage input source 102 and extract derived electrical indicators from the sampled voltage signal. The electrical telemetry module 110 may operate at a predetermined sampling frequency for continuously monitoring the input voltage supplied to computing devices within the system 100. The derived electrical indicators extracted by the electrical telemetry module 110 may include at least a rate of change of voltage and a short-term variance. The electrical telemetry module 110 processes the sampled voltage signal to characterize the current stability of the power envelope and identify early indicators of degradation, such as brownouts or brief voltage dips.

[0057] The predictive control engine 120 is communicatively coupled to the electrical telemetry module 110. The predictive control engine 120 receives the derived electrical indicators from the electrical telemetry module 110 in real time. Based on the derived electrical indicators, the predictive control engine 120 computes a predictive power stability index and determines a predicted power state. The predictive control engine 120 may implement a deterministic power-to-model mapping function that maps the predicted power state to a corresponding neural model configuration from a plurality of pre-optimized neural model configurations stored in the adaptive neural execution layer 130. The deterministic power-to-

model mapping function ensures that model reconfiguration decisions are repeatable and methodical rather than arbitrary.

[0058] The adaptive neural execution layer 130 is communicatively coupled to the predictive control engine 120. The adaptive neural execution layer 130 is configured to store a plurality of pre-optimized neural model configurations having different computational complexities. Based on the predicted power state received from the predictive control engine 120, the adaptive neural execution layer 130 dynamically selects and activates one of the plurality of pre-optimized neural model configurations to maintain continuous neural inference. The adaptive neural execution layer 130 produces an inference output 140 representing the results of neural network computation.

[0059] The system 100 operates as a closed-loop framework where voltage monitoring, predictive analysis, and model adaptation are continuously repeated. A feedback path returns operational status from the inference output 140 to the electrical telemetry module 110, thereby forming a closed-loop control mechanism. The system 100 establishes a direct cyber-physical control loop for changing edge intelligence from static deployment to predictive and energy-adaptive execution. The closed-loop integration of electrical telemetry and neural architecture control provides a cyber-physical feedback mechanism for energy-conscious artificial intelligence operation.

[0060] In operation, the electrical telemetry module 110 continuously monitors voltage characteristics and detects short-term predictive indicators such as voltage trend and voltage rate of change. The predictive control engine 120 analyzes the derived electrical indicators to determine whether power circumstances are stable, deteriorating, or unstable. When the predictive control engine 120 identifies a forecasted drop in available power, the predictive control engine 120 directs the adaptive neural execution layer 130 to dynamically modify the neural execution state by selecting a model configuration that is structurally lighter or employs reduced precision. The adaptation takes place dynamically without interfering with the inference process. When stable power conditions are re-established, the system 100 returns to a full-capability configuration. The continuous operation of the

closed-loop framework enables the system 100 to remain responsive to real-time changes in power conditions rather than reacting after degradation occurs.

[0061] FIG. 2 illustrates a state transition diagram of adaptive neural model switching governed by predicted power conditions according to various embodiments. The state transition diagram depicts how the system 100 dynamically transitions between different neural model configurations based on the predicted power state determined by the predictive control engine 120. As described above, the predictive control engine 120 determines a predicted power state based on the predictive power stability index, and the adaptive neural execution layer 130 stores a plurality of pre-optimized neural model configurations having different computational complexities and dynamically selects and activates one of the plurality of pre-optimized neural model configurations based on the predicted power state to maintain continuous neural inference.

[0062] The state transition diagram includes three distinct operational states corresponding to different power conditions. A full-precision model state 202 represents operation under stable power conditions where the system 100 executes a full-precision neural model configuration to deliver maximum inference accuracy. A quantized model state 204 represents operation under moderate power instability where the system 100 executes a reduced-precision quantized neural model configuration that balances accuracy and energy consumption. A lightweight model state 206 represents operation under severe or critical power conditions where the system 100 executes a computationally lightweight neural model configuration to minimize energy demands and reduce computing strain.

[0063] The plurality of pre-optimized neural model configurations may comprise a full-precision neural model configuration corresponding to the full-precision model state 202, a reduced-precision quantized neural model configuration corresponding to the quantized model state 204, and a computationally lightweight neural model configuration corresponding to the lightweight model state 206. Each configuration accomplishes the same inference objective but differs according to the degree of computational complexity, memory bandwidth requirements, and energy consumption.

[0064] The state transition diagram depicts three transitions between the operational states. A voltage drop predicted transition 208 transitions the system 100 from the full-precision model state 202 to the quantized model state 204 when the predictive control engine 120 predicts an impending voltage drop while operating in the full-precision model state 202. The voltage drop predicted transition 208 occurs when the predictive control engine 120 analyzes voltage trends and determines that power circumstances are deteriorating from stable conditions toward moderate instability.

[0065] A severe degradation predicted transition 210 transitions the system 100 from the quantized model state 204 to the lightweight model state 206 when the predictive control engine 120 predicts severe power degradation while operating in the quantized model state 204. The severe degradation predicted transition 210 occurs when the predictive control engine 120 determines that power circumstances are transitioning from moderate instability toward unstable or critical conditions.

[0066] A power restored transition 212 transitions the system 100 from the lightweight model state 206 back to the full-precision model state 202 when stable power conditions are re-established. The power restored transition 212 enables the system 100 to automatically recover to full-performance configuration mode with the restoration of stable voltage levels. The system 100 reduces computing strain under limited power conditions while enabling automated recovery after power envelope stabilization.

[0067] The predictive control engine 120 may apply hysteresis and state persistence constraints to prevent rapid oscillations between neural model configurations during fluctuating power conditions. The hysteresis and state persistence constraints prevent the system 100 from oscillating rapidly between different model configurations when power conditions hover near transition thresholds, thereby reducing switching-related latency and ensuring smoother inference operation in noisy or uncertain electrical environments.

[0068] In some cases, the predictive control engine 120 may implement alternative state classification approaches. In one variation, a binary classification scheme may classify power conditions as either stable or unstable, resulting in

transitions between two operational states. In another variation, a tri-state power classification scheme may classify power conditions as stable, deteriorating, or unstable, corresponding to the three operational states depicted in FIG. 2. In yet another variation, a multi-level classification scheme may classify power conditions across more than three discrete levels, enabling finer-grained adaptation with additional intermediate neural model configurations. The tri-state power classification depicted in FIG. 2 provides a balance between adaptation granularity and system complexity.

[0069] The state transitions depicted in FIG. 2 occur dynamically during runtime without halting the inference process. When the predictive control engine 120 identifies a forecasted drop in available power, the adaptive neural execution layer 130 dynamically modifies the neural execution state by selecting a model configuration that is structurally lighter or employs reduced precision. The adaptation takes place dynamically without interfering with the inference process. When stable power conditions are re-established, the system 100 returns to a full-capability configuration through the power restored transition 212.

[0070] FIG. 3 illustrates a detailed schematic of electrical telemetry module 110 and predictive control engine 120 according to various embodiments. As described above, the electrical telemetry module 110 is configured to continuously sample an input voltage signal from the voltage input source 102 and extract derived electrical indicators from the sampled voltage signal, wherein the derived electrical indicators include at least a rate of change of voltage and a short-term variance. The predictive control engine 120 receives the derived electrical indicators from the electrical telemetry module 110 and computes a predictive power stability index based on the derived electrical indicators.

[0071] The electrical telemetry module 110 comprises a voltage sampling unit 302, an instantaneous voltage magnitude calculator 304, a moving average computation unit 306, a rate-of-change extractor 308, and a short-term variance calculator 310. The voltage sampling unit 302 may continuously sample the input voltage signal from the voltage input source 102 at a predetermined sampling frequency. The instantaneous voltage magnitude calculator 304 may compute an

instantaneous voltage magnitude from each sampled voltage value. The moving average computation unit 306 may compute a moving average voltage over a sliding time window to filter transient noise and provide a smoothed representation of voltage trends. The rate-of-change extractor 308 may compute a rate of change of voltage (dV/dt) representing the temporal derivative of the voltage signal. The short-term variance calculator 310 may compute a short-term variance representing the statistical dispersion of voltage values within a recent time window.

[0072] The derived electrical indicators extracted by the electrical telemetry module 110 may further include an instantaneous voltage magnitude and a moving average voltage computed over a sliding time window. In some cases, the derived electrical indicators may include fluctuation intensity as an additional metric for characterizing the current stability of the power envelope. The derived electrical indicators may be provided to the predictive control engine 120 in real time for power state prediction.

[0073] The voltage sampling unit 302, instantaneous voltage magnitude calculator 304, moving average computation unit 306, rate-of-change extractor 308, and short-term variance calculator 310 may be implemented using various computational approaches. In some cases, the computational units may comprise dedicated hardware accelerators configured to perform the respective calculations with low latency and reduced power consumption. In some cases, the computational units may comprise software modules executing on a general-purpose processor or microcontroller. In some cases, the computational units may comprise hybrid implementations combining hardware acceleration for time-sensitive operations with software modules for configuration and control functions.

[0074] The predictive control engine 120 comprises a long short-term memory (LSTM) model 320, a predictive power stability index generator 312, a normalized score generator 314, a power-to-model mapping function 316, and a hysteresis module 318. The LSTM model 320 may be configured to analyze a time-series sequence of voltage readings and classify a current power condition as one of a plurality of power states. The LSTM model 320 may receive a short time-series sequence of recent voltage readings as input for learning temporal dependencies

and identifying trends in the voltage signal. Based on the analysis, the LSTM model 320 may predict the current power condition as low, medium, or high, thereby enabling proactive anticipation of power degradation. In some cases, the LSTM model 320 may achieve a final test accuracy of 90% for predicting power states from voltage sequences, demonstrating effective prediction of power states.

[0075] The predictive power stability index generator 312 may compute a predictive power stability index based on the derived electrical indicators received from the electrical telemetry module 110. The predictive power stability index may comprise a weighted combination of a moving average voltage, the rate of change of voltage, and the short-term variance. The predictive power stability index may express the probability of impending power degradation within a specified prediction horizon. The predictive control engine 120 may use a short-term prediction horizon to forecast whether voltage will fall below a set safety threshold.

[0076] The normalized score generator 314 may generate a normalized predictive power stability score for uniform interpretation across different hardware platforms and voltage ranges. The normalized predictive power stability score may abstract away hardware-specific voltage characteristics, allowing the same power-to-model mapping logic to be applied uniformly regardless of the underlying hardware platform or nominal voltage range.

[0077] The power-to-model mapping function 316 may implement a deterministic power-to-model mapping function that maps the predicted power state to a corresponding neural model configuration from the plurality of pre-optimized neural model configurations. The power-to-model mapping function 316 may be configurable and tunable based on application-specific restrictions including latency tolerance, accuracy requirements, and energy budgets. The deterministic nature of the power-to-model mapping function 316 ensures that model reconfiguration decisions are repeatable and methodical rather than arbitrary.

[0078] The hysteresis module 318 may apply hysteresis and state persistence constraints to prevent rapid oscillations between neural model configurations during fluctuating power conditions. The hysteresis module 318 may enable stability when selecting a model in noisy or uncertain electrical environments by

preventing the system 100 from oscillating rapidly between different model configurations when power conditions hover near transition thresholds.

[0079] In an operational flow example, an edge device may be placed in an environment with an electricity supply that fluctuates. The voltage sampling unit 302 may sample a voltage reading of 11.8V with a negative rate of change (dV/dt = -0.15 V/sec), indicating a declining trend in the data from the electrical telemetry module 110. The predictive control engine 120 may analyze the trend using the short-term prediction horizon and determine a forecast that the voltage reading will continue to fall below a set safety threshold of 11.2V. Based on the forecast, the predictive control engine 120 may deem the system 100 to be in a degraded state and select a reduced precision neural execution configuration through the power-to-model mapping function 316.

[0080] FIG. 4 illustrates a schematic of adaptive neural execution layer 130 with a pre-optimized model ensemble according to various embodiments. As described above, the adaptive neural execution layer 130 is communicatively coupled to the predictive control engine 120 and is configured to store a plurality of pre-optimized neural model configurations having different computational complexities and dynamically select and activate one of the plurality of pre-optimized neural model configurations based on the predicted power state to maintain continuous neural inference.

[0081] The adaptive neural execution layer 130 comprises a model selector 408, a full-precision neural model configuration 402, a quantized neural model configuration 404, a lightweight neural model configuration 406, a state retention mechanism 410, and a preloaded memory store 412. The model selector 408 receives the predicted power state from the predictive control engine 120 and dynamically activates one of the full-precision neural model configuration 402, the quantized neural model configuration 404, or the lightweight neural model configuration 406 to produce an inference output 140.

[0082] The full-precision neural model configuration 402 may comprise a neural network model that operates with full floating-point precision weights and activations. The full-precision neural model configuration 402 may deliver

maximum inference accuracy when power conditions are stable and abundant. In some cases, the full-precision neural model configuration 402 may utilize 32-bit floating-point representations for weights and intermediate computations.

[0083] The quantized neural model configuration 404 may comprise a neural network model that operates with reduced numerical precision. In some cases, the quantized neural model configuration 404 may employ 8-bit integer quantization for weights and activations. In some cases, the quantized neural model configuration 404 may employ 4-bit quantization or mixed-precision quantization schemes where different layers utilize different bit-widths based on sensitivity analysis. The quantized neural model configuration 404 may offer a balance between inference accuracy and energy consumption during moderate power instability.

[0084] The lightweight neural model configuration 406 may comprise a neural network model with reduced computational complexity. In some cases, the lightweight neural model configuration 406 may be generated through pruning strategies that remove redundant weights or neurons from a larger model. In some cases, the lightweight neural model configuration 406 may be generated through knowledge distillation techniques where a smaller student model learns to approximate the behavior of a larger teacher model. In some cases, the lightweight neural model configuration 406 may employ structured pruning that removes entire channels or layers, unstructured pruning that removes individual weights, or a combination of pruning approaches. In some cases, the lightweight neural model configuration 406 may utilize depth-wise separable convolutions, bottleneck architectures, or other efficient neural network design patterns. The lightweight neural model configuration 406 may enable continued inference operation with reduced energy demands during severe power constraints.

[0085] The preloaded memory store 412 may store the full-precision neural model configuration 402, the quantized neural model configuration 404, and the lightweight neural model configuration 406 as a model ensemble. The neural model configurations may be stored in the preloaded memory store 412 prior to runtime, enabling low-latency switching without on-the-fly restructure or retraining. The

preloaded memory store 412 may comprise random access memory (RAM), flash memory, or other storage media accessible with low read latency. By preloading the neural model configurations in memory, the adaptive neural execution layer 130 may perform low-latency switching between the plurality of pre-optimized neural model configurations. The model ensemble storage approach enables the system 100 to transition between neural model configurations without incurring latency associated with loading models from external storage or performing runtime model compression.

[0086] The model selector 408 may receive the predicted power state from the predictive control engine 120 and select a corresponding neural model configuration from the preloaded memory store 412. The model selector 408 may activate the full-precision neural model configuration 402 when the predicted power state indicates stable power conditions. The model selector 408 may activate the quantized neural model configuration 404 when the predicted power state indicates moderate power instability. The model selector 408 may activate the lightweight neural model configuration 406 when the predicted power state indicates severe or critical power conditions. The model selector 408 may implement memory-aware scheduling to reduce transition overhead so that model change does not cause noticeable latency spikes or inference discontinuities.

[0087] The state retention mechanism 410 may maintain continuity of an intermediate inference context during switching between neural model configurations. The state retention mechanism 410 may preserve computational progress and contextual information when transitioning between the full-precision neural model configuration 402, the quantized neural model configuration 404, and the lightweight neural model configuration 406. In some cases, the state retention mechanism 410 may store hidden states, feature maps, or other intermediate representations that enable the system 100 to resume inference from an intermediate state rather than restarting from the beginning. The state retention mechanism 410 may reduce redundant computation and maintain inference coherence across model transitions.

[0088] In some cases, the plurality of pre-optimized neural model configurations may include additional intermediate configurations beyond the three configurations depicted in FIG. 4. In some cases, the plurality of pre-optimized neural model configurations may include multiple quantized configurations with different bit-widths, such as 16-bit, 8-bit, and 4-bit quantized models. In some cases, the plurality of pre-optimized neural model configurations may include models generated through different compression techniques applied in combination, such as quantization combined with pruning or knowledge distillation combined with quantization.

[0089] In operation, when the model selector 408 receives a predicted power state indicating a transition from stable to deteriorating power conditions, the model selector 408 may deactivate the full-precision neural model configuration 402 and activate the quantized neural model configuration 404 from the preloaded memory store 412. The state retention mechanism 410 may transfer relevant intermediate inference context from the full-precision neural model configuration 402 to the quantized neural model configuration 404 to maintain inference continuity. The quantized neural model configuration 404 may then continue producing inference output 140 with reduced energy consumption. When stable power conditions are re-established, the model selector 408 may activate the full-precision neural model configuration 402 to restore full inference accuracy.

[0090] FIG. 5 illustrates a flowchart of a method 500 for predictive power-aware neural inference according to various embodiments. The method 500 may be performed by the system 100 described above with reference to FIGS. 1-4. The method 500 comprises a sequence of steps that enable continuous, closed-loop adaptation of neural inference operations in response to predicted power conditions.

[0091] The method 500 begins with a step of sampling voltage signal 502. At the step of sampling voltage signal 502, the electrical telemetry module 110 may continuously sample an input voltage signal from the voltage input source 102. The voltage sampling may occur at a predetermined sampling frequency to capture voltage fluctuations with sufficient temporal resolution. The sampled voltage signal provides raw data for subsequent analysis of power supply characteristics.

[0092] Following the step of sampling voltage signal 502, the method 500 proceeds to a step of extracting derived electrical indicators 504. At the step of extracting derived electrical indicators 504, the electrical telemetry module 110 may process the sampled voltage signal to extract multiple derived electrical indicators. The derived electrical indicators may include an instantaneous voltage magnitude representing the current voltage level, a moving average voltage computed over a sliding time window to filter transient noise, a rate of change of voltage (dV/dt) representing the temporal derivative of the voltage signal, and a short-term variance representing the statistical dispersion of voltage values within a recent time window. The derived electrical indicators characterize the current stability of the power envelope and provide inputs for power state prediction.

[0093] The method 500 then proceeds to a step of computing predictive power stability index 506. At the step of computing predictive power stability index 506, the predictive control engine 120 may compute a predictive power stability index based on the derived electrical indicators received from the electrical telemetry module 110. The predictive power stability index may comprise a weighted combination of the moving average voltage, the rate of change of voltage, and the short-term variance. The predictive power stability index may express the probability of impending power degradation within a specified prediction horizon. In some cases, the predictive control engine 120 may generate a normalized predictive power stability score for uniform interpretation across different hardware platforms and voltage ranges.

[0094] Following the step of computing predictive power stability index 506, the method 500 proceeds to a step of determining predicted power state 508. At the step of determining predicted power state 508, the predictive control engine 120 may determine a predicted power state based on the predictive power stability index. The predicted power state may indicate whether power conditions are stable, deteriorating, or unstable. In some cases, the predictive control engine 120 may employ a long short-term memory (LSTM) model to analyze a time-series sequence of voltage readings and classify a current power condition as one of a plurality of power states.

[0095] The method 500 then proceeds to a decision step evaluating power state 510. At the decision step evaluating power state 510, the method 500 evaluates the predicted power state and routes execution to one of multiple model selection steps based on the evaluation result. The decision step evaluating power state 510 may implement various decision logic approaches. In some cases, the decision step evaluating power state 510 may employ threshold-based rules that compare the predictive power stability index against predefined threshold values to determine the appropriate model selection. In some cases, the decision step evaluating power state 510 may employ machine learning classifiers trained to map power state indicators to model selection decisions. In some cases, the decision step evaluating power state 510 may employ fuzzy logic controllers that handle uncertainty and gradual transitions between power states using membership functions and fuzzy inference rules. In some cases, the decision step evaluating power state 510 may employ other decision mechanisms including rule-based expert systems, decision trees, or ensemble methods that combine multiple decision approaches.

[0096] Based on the evaluation at the decision step evaluating power state 510, the method 500 routes execution to one of three model selection steps. When the predicted power state indicates stable power conditions, the method 500 proceeds to a step of selecting full-precision model 512. At the step of selecting full-precision model 512, the adaptive neural execution layer 130 may select and activate the full-precision neural model configuration 402 to deliver maximum inference accuracy.

[0097] When the predicted power state indicates moderate power instability, the method 500 proceeds to a step of selecting quantized model 514. At the step of selecting quantized model 514, the adaptive neural execution layer 130 may select and activate the quantized neural model configuration 404 to balance inference accuracy and energy consumption.

[0098] When the predicted power state indicates severe or critical power conditions, the method 500 proceeds to a step of selecting lightweight model 516. At the step of selecting lightweight model 516, the adaptive neural execution layer 130 may select and activate the lightweight neural model configuration 406 to minimize energy demands and reduce computing strain.

[0099] Following the model selection at the step of selecting full-precision model 512, the step of selecting quantized model 514, or the step of selecting lightweight model 516, the method 500 proceeds to a step of executing neural inference 518. At the step of executing neural inference 518, the adaptive neural execution layer 130 may execute neural inference using the selected neural model configuration to produce an inference output 140. The step of executing neural inference 518 may utilize the state retention mechanism 410 to maintain continuity of an intermediate inference context when the selected model configuration differs from a previously active model configuration.

[0100] Following the step of executing neural inference 518, the method 500 proceeds through a feedback loop 520 that returns to the step of sampling voltage signal 502. The feedback loop 520 enables continuous closed-loop operation where voltage monitoring, predictive analysis, and model adaptation are continuously repeated. The feedback loop 520 may return operational status from the inference output 140 to the electrical telemetry module 110, thereby forming a closed-loop control mechanism. The continuous operation of the feedback loop 520 enables the system 100 to remain responsive to real-time changes in power conditions rather than reacting after degradation occurs.

[0101] The method 500 operates as a continuous closed-loop process where the steps are repeated iteratively during runtime. The closed-loop operation enables proactive adaptation of neural inference operations in response to anticipated power conditions. The method 500 may apply hysteresis and state persistence constraints during the decision step evaluating power state 510 to prevent rapid oscillations between neural model configurations during fluctuating power conditions.

[0102] In some cases, experimental validation of the method 500 may be performed using a controlled voltage simulation environment. The controlled voltage simulation environment may generate dynamic voltage values representing unstable power scenarios for testing and validation purposes. The voltage simulation environment may simulate various power conditions including stable voltage, gradual voltage decline, sudden voltage drops, voltage fluctuations, and voltage recovery patterns. The voltage simulation environment may enable

evaluation of the method 500 under controlled conditions that replicate real-world power instability scenarios encountered in edge computing deployments.

[0103] It is to be appreciated that the concepts, systems, and techniques sought to be protected herein are not limited to use in the example applications described herein, but rather may be useful in substantially any application where it is desired to maintain continuous neural inference operations under variable or unstable power supply conditions. The system 100 for predictive power-aware neural inference is susceptible of industrial application across a wide range of industries and deployment scenarios.

[0104] In the field of industrial automation, the system 100 may be deployed in manufacturing facilities where edge computing devices perform real-time quality inspection, predictive maintenance, or process monitoring. Such industrial environments may experience power fluctuations due to heavy machinery operation, and the adaptive neural execution capabilities disclosed herein may enable uninterrupted AI-driven monitoring even during brownouts or voltage instability.

[0105] In the field of smart agriculture, the system 100 may be deployed in remote agricultural monitoring stations powered by solar panels or wind turbines. The predictive power-aware capabilities may enable continuous crop health analysis, pest detection, or irrigation control even when renewable energy sources provide intermittent or variable power supply.

[0106] In the field of autonomous vehicles and transportation, the system 100 may be deployed in electric vehicles, drones, or unmanned ground vehicles where battery depletion or power system anomalies may affect available computational resources. The adaptive model switching capabilities may enable continued perception, navigation, or decision-making functions during power-constrained operating conditions.

[0107] In the field of healthcare and medical devices, the system 100 may be deployed in wearable health monitors, portable diagnostic equipment, or remote patient monitoring systems. Such devices may operate on battery power with

limited capacity, and the predictive adaptation capabilities may enable continuous health monitoring and anomaly detection while extending battery life.

[0108] In the field of smart infrastructure and building automation, the system 100 may be deployed in smart building systems, environmental monitoring stations, or distributed sensor networks. The adaptive neural inference capabilities may enable continuous occupancy detection, energy management, or security monitoring even during grid instability or backup power operation.

[0109] In the field of telecommunications and network infrastructure, the system 100 may be deployed in edge computing nodes, base stations, or network gateways located in areas with unreliable power infrastructure. The predictive power-aware capabilities may enable continued network traffic analysis, content caching, or edge AI services during power fluctuations.

[0110] While particular embodiments and applications of the present disclosure have been illustrated and described in connection with edge computing and power-constrained environments, it is to be understood that embodiments of the disclosure are not limited to the precise constructions and compositions disclosed herein and that various modifications, changes, and variations can be apparent from the foregoing descriptions without departing from the spirit and scope of the disclosure as defined in the appended claims.

[0111] There are several technical effects of the system for predictive power-aware neural inference of the present disclosure. First, the integration of an electrical telemetry module that continuously samples voltage signals and extracts derived electrical indicators including rate of change and short-term variance enables proactive detection of impending power instability before degradation occurs, thereby allowing anticipatory adaptation rather than reactive responses to power failures. Second, the computation of a predictive power stability index based on multiple derived electrical indicators provides a quantitative and systematic basis for model selection decisions, enabling repeatable and methodical reconfiguration across varying power conditions. Third, the adaptive neural execution layer with pre-optimized neural model configurations of different computational complexities allows the system to maintain continuous inference

operation even during brownouts or voltage fluctuations by dynamically selecting between full-precision, quantized, and lightweight configurations matched to current power availability. Fourth, the preloading of neural model configurations in memory enables low-latency switching between configurations, avoiding delays associated with loading models from external storage during power state transitions and thereby maintaining inference continuity. Fifth, the state retention mechanism preserves intermediate inference context during model transitions, reducing redundant computation and maintaining inference coherence when switching between neural model configurations. Sixth, the closed-loop feedback architecture integrating electrical telemetry with neural architecture control establishes a cyber-physical control mechanism that transforms edge intelligence from static deployment to predictive and energy-adaptive execution. Seventh, the hysteresis and state persistence constraints prevent rapid oscillations between neural model configurations when power conditions hover near transition thresholds, reducing switching-related latency and ensuring smoother inference operation in noisy or uncertain electrical environments. Eighth, the normalized predictive power stability score enables portability and consistent behavior across diverse hardware platforms and voltage ranges, allowing uniform application of power-to-model mapping logic regardless of deployment environment. These technical effects collectively contribute to improved operational resilience and durability in electrically unstable edge computing environments.

[0112] It will be appreciated that the above-detailed description, along with FIGs, is provided to illustrate the architecture and operation of exemplary embodiments of the present invention. Various modifications and enhancements can be made without departing from the scope of the invention. All such variations, as would be recognized by those skilled in the art, are intended to be within the scope of the present disclosure, which is defined by the claims that follow.

[0113] It will be appreciated by persons skilled in the art that while the invention has been described with reference to specific embodiments and accompanying drawings, numerous modifications, substitutions, variations, and equivalents are possible without departing from the spirit or scope of the present

invention. The use of singular terms such as “a,” “an,” and “the” is not intended to limit the disclosed elements to a single instance unless explicitly stated, and such terms should be interpreted to include plural forms as applicable. Likewise, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having,” and their grammatical variants are intended to be open-ended and non-limiting, and should be interpreted as meaning “including but not limited to.” Any enumerated listing of components, features, or elements does not imply that the items are mutually exclusive unless expressly stated.

[0114] It is further understood that where features, characteristics, or elements are described in connection with a Markush group or a disjunctive phrase (e.g., “A or B”), the invention includes all possible combinations and sub-combinations thereof unless explicitly excluded. Thus, a statement referring to “at least one of A, B, and C” should be interpreted to mean any one or more of A, B, and C, individually or in any combination. The language used in this specification is selected primarily for clarity and illustrative purposes and should not be construed to limit the inventive scope unless specifically recited in the claims. Accordingly, the foregoing description of embodiments should be regarded as illustrative and not restrictive, with the scope of the invention being defined solely by the appended claims and their legal equivalents

[0115] It will be recognized that various features, elements, and combinations thereof described herein may be desirably adapted for alternative implementations or other applications. Many such modifications, substitutions, enhancements, or equivalents may become apparent to those skilled in the art upon reading this disclosure and are considered to fall within the scope and spirit of the present invention. The invention is not limited to the precise configurations or exemplary embodiments set forth in this specification, and various alternatives may be used without departing from the intended objectives. The claims, and not the detailed description, shall define the legal scope of protection afforded by the present disclosure.

[0116] In interpreting the specification and claims, all terms should be construed in the broadest reasonable manner consistent with the context and the

understanding of those skilled in the art. The use of terms such as “comprises,” “comprising,” “includes,” “including,” or “has” should be interpreted to be non-exclusive and not limited to the stated elements alone. Moreover, references to “one embodiment,” “an embodiment,” or “in certain embodiments” are not meant to imply that such features are required or exclusive, and any feature described in connection with one embodiment may be used in combination with other features or embodiments unless clearly prohibited or contextually inconsistent.

[0117] It should be further appreciated that reference throughout the specification to features, operations, or components using singular terms should not be construed as excluding a plurality thereof unless the context expressly indicates otherwise. Similarly, features described in the context of grouped or listed elements (e.g., A, B, and C) should be understood to encompass individual, multiple, or all combinations of said elements. Where ranges are given, all intermediate values and subranges are understood to be disclosed as if specifically recited. The scope of the invention should therefore be construed as inclusive of all such combinations and logical extensions as would be appreciated by a person of ordinary skill in the relevant technical field

[0118] In another embodiment, the above disclosure is a description of the invention and is not intended to limit the scope of the invention. Other variations and modifications of the above-described embodiment shall be apparent to those skilled in the art and are intended to fall within the scope of the invention as defined in the following claims.

Dated This ..... day of May 2026

Kuldeep

(Kuldeep Singh)

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Indian Patent Agent Regn No. IN/PA-4358

CLAIMS

WE CLAIM:

1. A system (100) for predictive power-aware neural inference, comprising:

an electrical telemetry module (110) configured to continuously sample an input voltage signal from a voltage input source (102) and extract derived electrical indicators from the sampled voltage signal, wherein the derived electrical indicators include at least a rate of change of voltage and a short-term variance;

a predictive control engine (120) communicatively coupled to the electrical telemetry module (110), the predictive control engine (120) configured to:

receive the derived electrical indicators from the electrical telemetry module (110);

compute a predictive power stability index based on the derived electrical indicators; and

determine a predicted power state based on the predictive power stability index; and

an adaptive neural execution layer (130) communicatively coupled to the predictive control engine (120), the adaptive neural execution layer (130) configured to:

store a plurality of pre-optimized neural model configurations having different computational complexities; and

dynamically select and activate one of the plurality of pre-optimized neural model configurations based on the predicted power state to maintain continuous neural inference.

2. The system (100) of claim 1, wherein the derived electrical indicators further include an instantaneous voltage magnitude and a moving average voltage computed over a sliding time window.

3. The system (100) of claim 1, wherein the plurality of pre-optimized neural model configurations comprise:

a full-precision neural model configuration;

a reduced-precision quantized neural model configuration; and

a computationally lightweight neural model configuration.

4. The system (100) of claim 1, wherein the predictive control engine (120) is further configured to implement a deterministic power-to-model mapping function that maps the predicted power state to a corresponding neural model configuration from the plurality of pre-optimized neural model configurations.

5. The system (100) of claim 1, wherein the predictive control engine (120) is further configured to apply hysteresis and state persistence constraints to prevent rapid oscillations between neural model configurations during fluctuating power conditions.

6. The system (100) of claim 1, wherein the predictive control engine (120) comprises a long short-term memory (LSTM) model configured to analyze a time-series sequence of voltage readings and classify a current power condition as one of a plurality of power states.

7. The system (100) of claim 1, wherein the adaptive neural execution layer (130) is further configured to perform low-latency switching between the plurality of pre-optimized neural model configurations by preloading the neural model configurations in memory.

8. The system (100) of claim 1, wherein the predictive power stability index comprises a weighted combination of a moving average voltage, the rate of change of voltage, and the short-term variance.

9. The system (100) of claim 1, wherein the adaptive neural execution layer (130) is further configured to implement a state retention mechanism to maintain continuity of an intermediate inference context during switching between neural model configurations.

10. The system (100) of claim 1, wherein the predictive control engine (120) is further configured to generate a normalized predictive power stability score for uniform interpretation across different hardware platforms and voltage ranges.

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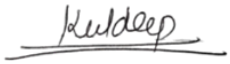
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ABSTRACT

SYSTEM FOR PREDICTIVE POWER-AWARE ADAPTIVE NEURAL INFERENCE

A system (100) for predictive power-aware neural inference comprises an electrical telemetry module (110) configured to continuously sample an input voltage signal from a voltage input source (102) and extract derived electrical indicators including at least a rate of change of voltage and a short-term variance. A predictive control engine (120) communicatively coupled to the electrical telemetry module (110) receives the derived electrical indicators, computes a predictive power stability index based on the derived electrical indicators, and determines a predicted power state based on the predictive power stability index. An adaptive neural execution layer (130) communicatively coupled to the predictive control engine (120) stores a plurality of pre-optimized neural model configurations having different computational complexities and dynamically selects and activates one of the pre-optimized neural model configurations based on the predicted power state to maintain continuous neural inference.

Dated This ..... day of May 2026



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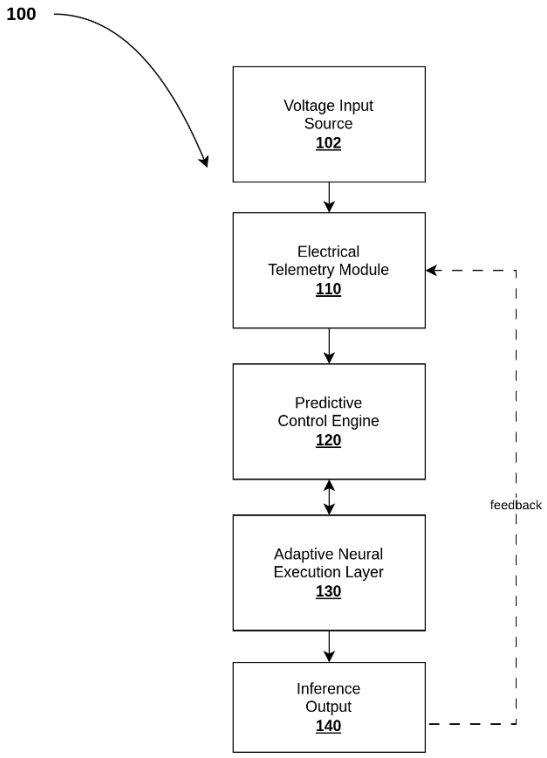


FIG. 1

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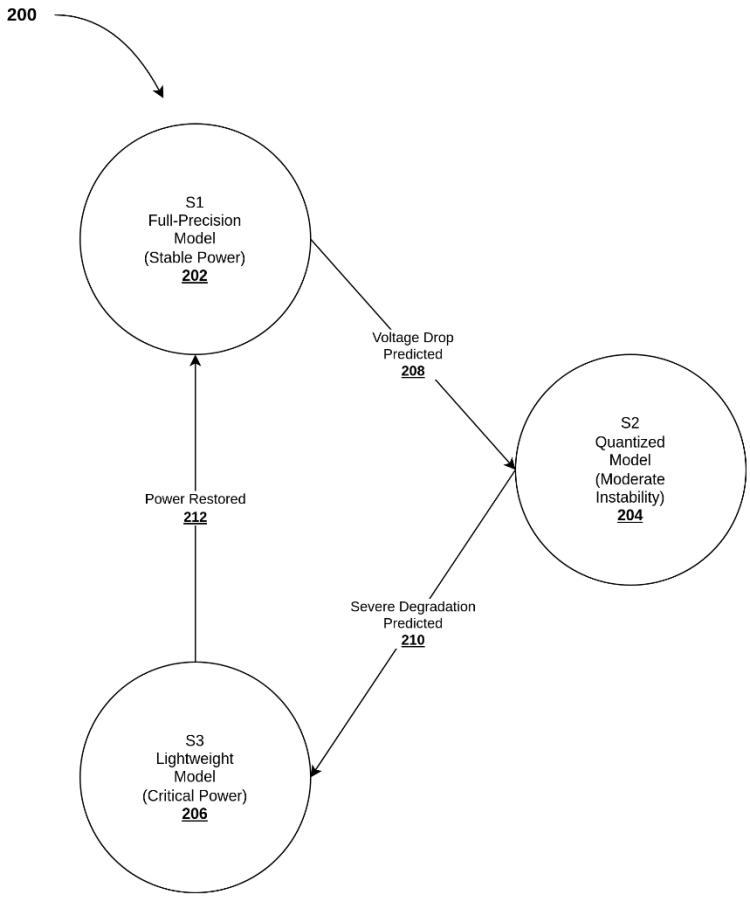


FIG. 2

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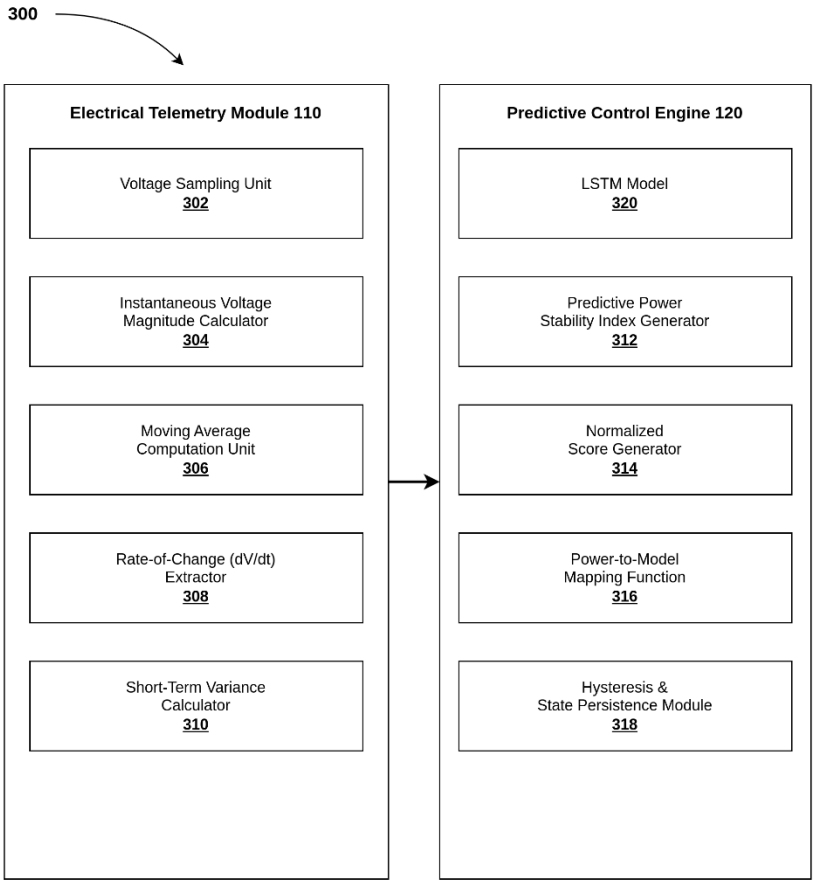


FIG. 3

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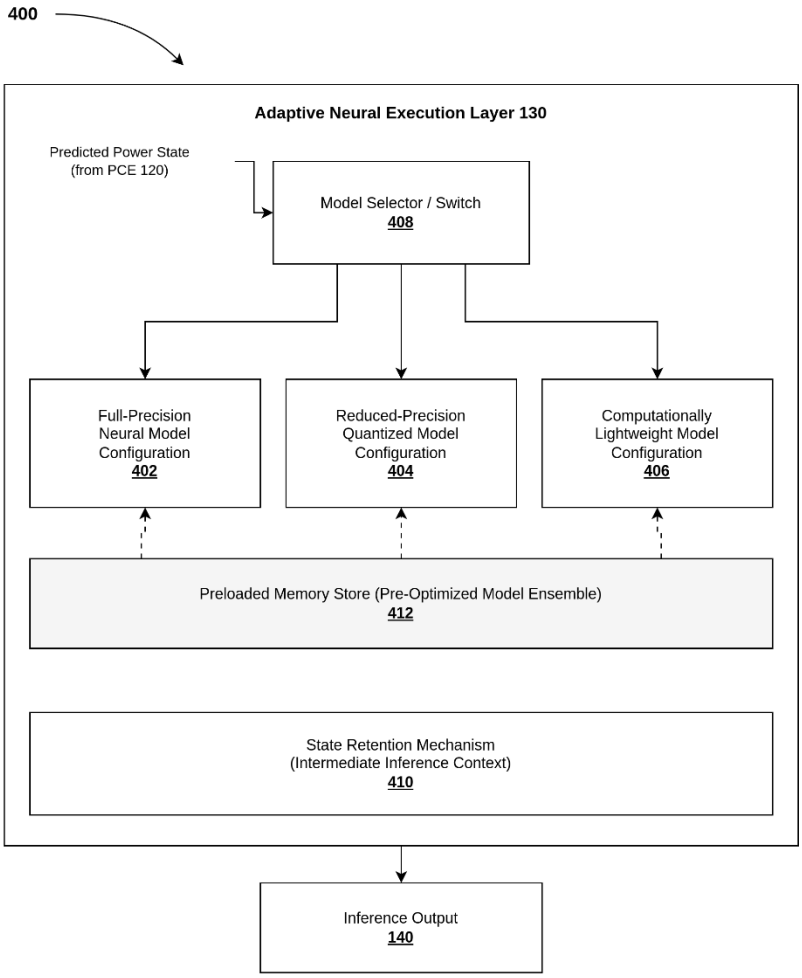


FIG. 4

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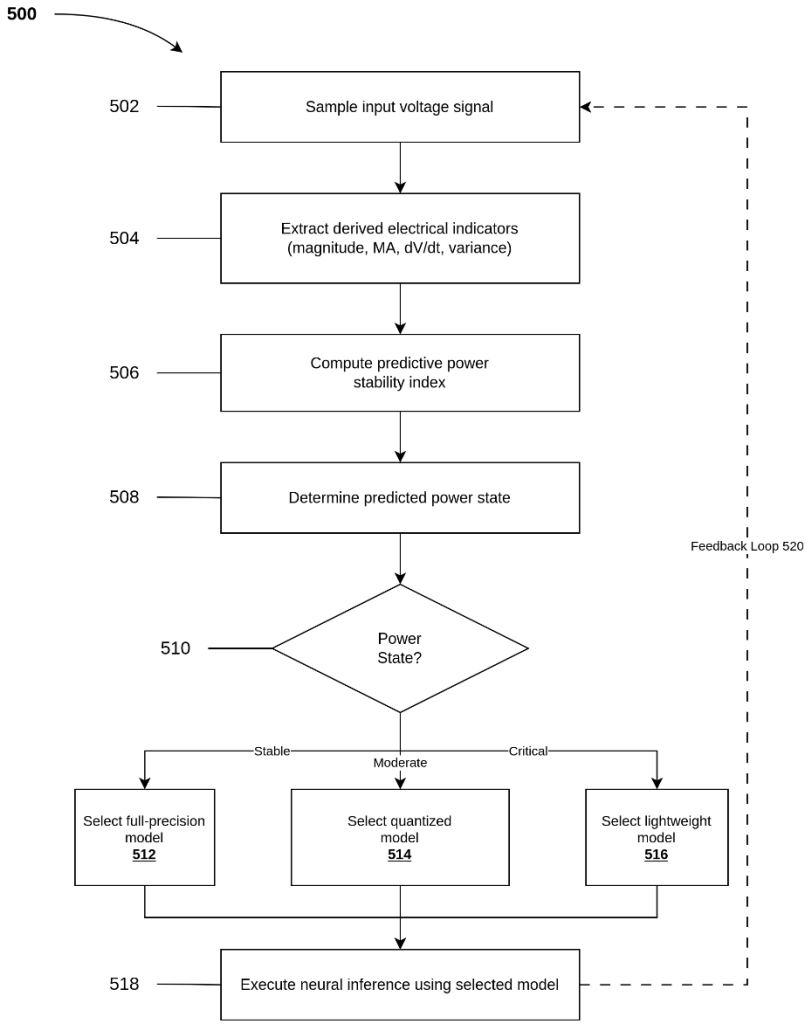


FIG. 5

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